

AP CSP | Lesson 17: Lists & Abstraction

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5: select exactly TWO. AP list indices start at 1. Indexed operations use 1 through current LENGTH; invalid indices cause an error. All examples use simple lists.

1. After $a \leftarrow 4$; $b \leftarrow a$; $a \leftarrow 9$, what is b ? [1 mark]

- A. 4 B. 9 C. 13 D. 0

Answer: _____

2. For $a \leftarrow [12, 7, 20]$, what is $a[2]$? [1 mark]

- A. 12 B. 7 C. 20 D. 2

Answer: _____

3. Starting with $a \leftarrow [12, 7, 20]$, execute $a[2] \leftarrow 9$. What is a ? [1 mark]

- A. $[12, 9, 7, 20]$ B. $[9, 7, 20]$ C. $[12, 9, 20]$ D. $[12, 7, 20, 9]$

Answer: _____

4. After $a \leftarrow [8, 5]$; $\text{REMOVE}(a, 1)$, which access is valid? [1 mark]

- A. $a[0]$ B. $a[2]$ C. $a[5]$ D. $a[1]$

Answer: _____

5. Which TWO statements are true? Select exactly TWO answers. [1 mark]

- A. $\text{LENGTH}([])$ is 0. B. $\text{REMOVE}(a, 2)$ searches for the value 2.
C. APPEND increases length by 1. D. A list cannot contain duplicate values.

Answer: _____

6. Why can a list of session minutes manage program complexity? [1 mark]

- A. It guarantees faster execution. B. One named collection can hold changing numbers of related entries.
C. It guarantees every entry is correct. D. It removes the need to check indices.

Answer: _____

AP CSP | Lesson 17: Lists & Abstraction

Written responses | Questions 7-9

7. [4 marks]

For $a \leftarrow [5, 5, 12, 8]$, give (a) $a[1]$, (b) $a[3]$, (c) $\text{LENGTH}(a)$, and (d) all valid indices.

8. [4 marks]

Start with $a \leftarrow [10, 20, 30]$. Give the full list after each step in order:

- (a) $\text{APPEND}(a, 40)$
- (b) $\text{INSERT}(a, 2, 15)$
- (c) $\text{REMOVE}(a, 4)$
- (d) $a[1] \leftarrow 5$

9. [4 marks]

Each case starts independently with $a \leftarrow [4, 8, 12]$. Give the final list after (a) $a[2] \leftarrow 6$, (b) $\text{INSERT}(a, 2, 6)$, (c) $\text{APPEND}(a, 6)$, and (d) $\text{REMOVE}(a, 2)$.

AP CSP | Lesson 17: Lists & Abstraction

Written responses | Questions 10-12

10. [4 marks]

Write four statements: (a) create sessions as [15, 30]; (b) append 45; (c) change the first value to 20; (d) display the current last element using LENGTH.

11. [4 marks]

Trace separately: `a ← []`; `APPEND(a, 30)`. Give (a) final `a`, (b) `LENGTH(a)`, (c) `a[1]`, and (d) what happens if `a[0]` is then accessed.

12. [4 marks]

A program stores session minutes in separate variables `s1,s2,s3`. (a) Propose a meaningful list name. (b) Give a list literal for 10, 20, 30. (c) Explain how an extra session can be added without inventing `s4`. (d) State one requirement that a list alone does not guarantee.
